



**EVALUATING WATER QUALITY USING THE HABITAT INTEGRITY INDEX AND
PHYSICOCHEMICAL PARAMETERS: A CASE STUDY IN THE PINDARÉ RIVER BASIN, MA**

**AVALIAÇÃO DA QUALIDADE DA ÁGUA UTILIZANDO O ÍNDICE DE INTEGRIDADE DO
HABITAT E PARÂMETROS FÍSICO-QUÍMICOS: UM ESTUDO DE CASO NA BACIA
HIDROGRÁFICA DO RIO PINDARÉ, MA**

**EVALUACIÓN DE LA CALIDAD DEL AGUA MEDIANTE EL ÍNDICE DE INTEGRIDAD DEL
HÁBITAT Y PARÁMETROS FÍSICOQUÍMICOS: UN ESTUDIO DE CASO EN LA CUENCA
DEL RÍO PINDARÉ, MA**

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ABSTRACT

This study aimed to evaluate the conservation status of 27 streams within the Pindaré River basin, Maranhão, Brazil. The assessment was conducted using the Habitat Integrity Index (HII), complemented by several physicochemical variables, including conductivity, turbidity, dissolved oxygen concentration, pH, temperature, depth, width, and canopy cover. Based on the HII, streams were categorized as preserved, altered, or degraded. The majority of the sampled streams were classified as altered or degraded, mainly due to riparian vegetation removal, erosion, and sedimentation processes. Preserved streams exhibited higher canopy cover, greater structural complexity, and more stable habitats, whereas degraded streams showed elevated water temperature and turbidity levels. Moreover, dissolved oxygen concentration and canopy cover were positively correlated with HII values, reinforcing their role as key indicators of habitat quality. The results highlight the effectiveness of the HII as a practical and low-cost tool for assessing environmental conditions in tropical streams. They also underscore the urgent need for riparian restoration and management strategies to mitigate anthropogenic impacts across the Amazonian region.

KEYWORDS: Rapid assessment protocol. Riparian forest. Water quality. Tropical stream. Aquatic ecosystem conservation.

RESUMO

Este estudo teve como objetivo avaliar o estado de conservação de 27 córregos na bacia do rio Pindaré, Maranhão, Brasil. A avaliação foi realizada utilizando o Índice de Integridade do Habitat (IIH), complementado por diversas variáveis físico-químicas, incluindo condutividade, turbidez, concentração de oxigênio dissolvido, pH, temperatura, profundidade, largura e cobertura vegetal. Com base no IIH, os córregos foram categorizados como preservados, alterados ou degradados. A maioria dos córregos amostrados foi classificada como alterada ou degradada, principalmente devido à remoção da vegetação ripária, erosão e processos de sedimentação.

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Os córregos preservados apresentaram maior cobertura vegetal, maior complexidade estrutural e habitats mais estáveis, enquanto os córregos degradados apresentaram níveis elevados de temperatura da água e turbidez. Além disso, a concentração de oxigênio dissolvido e a cobertura vegetal correlacionaram-se positivamente com os valores do IIH, reforçando seu papel como indicadores-chave da qualidade do habitat. Os resultados destacam a eficácia do IIH como uma ferramenta prática e de baixo custo para avaliar as condições ambientais em riachos tropicais. Eles também ressaltam a necessidade urgente de estratégias de restauração e manejo ripário para mitigar os impactos antropogênicos em toda a região amazônica.

PALAVRAS-CHAVE: Protocolo de avaliação rápida. Mata ciliar. Qualidade da água. Riacho tropical. Conservação de ecossistemas aquáticos.

RESUMEN

Este estudio tuvo como objetivo evaluar el estado de conservación de 27 arroyos en la cuenca del río Pindaré, Maranhão, Brasil. La evaluación se llevó a cabo mediante el Índice de Integridad del Hábitat (IIH), complementado por diversas variables fisicoquímicas, que incluyen conductividad, turbidez, concentración de oxígeno disuelto, pH, temperatura, profundidad, anchura y cobertura del dosel. Con base en el IIH, los arroyos fueron categorizados como preservados, alterados o degradados. La mayoría de los arroyos muestreados se clasificaron como alterados o degradados, principalmente debido a la eliminación de la vegetación riparia, la erosión y los procesos de sedimentación. Los arroyos preservados presentaron mayor cobertura de dosel, mayor complejidad estructural y hábitats más estables, mientras que los arroyos degradados mostraron niveles elevados de temperatura del agua y turbidez. Además, la concentración de oxígeno disuelto y la cobertura del dosel se correlacionaron positivamente con los valores del IIH, reforzando su papel como indicadores clave de la calidad del hábitat. Los resultados destacan la eficacia del IIH como una herramienta práctica y de bajo costo para evaluar las condiciones ambientales en arroyos tropicales. También subrayan la urgente necesidad de estrategias de restauración y manejo ripario para mitigar los impactos antropogénicos en toda la región amazónica.

PALABRAS CLAVE: Protocolo de evaluación rápida. Bosque de galería. Calidad del agua. Arroyo tropical. Conservación de ecosistemas acuáticos.

INTRODUCTION

The Amazon Basin, which covers an area of approximately 6 million km² and encompasses over 1,100 tributaries, is the largest hydrographic basin globally and plays an essential role in maintaining climate balance and biodiversity (Santos *et al.*, 2021). In recent decades, unregulated human population growth and economic activity intensification have increased the pressure on its water resources, consequently affecting both aquatic ecosystems and the human populations that depend on them (Santos *et al.*, 2020).

The main drivers of changes in water quality and ecological integrity are deforestation, agricultural expansion, logging, and urbanization, which collectively suppress natural vegetation



and profoundly modify aquatic habitats (Brasil *et al.*, 2020; Shi *et al.*, 2017). Riparian forests play a crucial role in the conservation of biodiversity and ecological processes and function as protection zones for water bodies (Martins *et al.*, 2022).

These forests, also referred to as galleries or riparian vegetation, are plant formations along rivers, springs, lakes, and reservoirs legally recognized as Permanent Preservation Areas by the Brazilian Forest Code (Correia *et al.*, 2019). In addition to acting as essential ecological corridors for fauna and flora connectivity, these maintain geological stability, control erosion, filter sediments, and regulate the aquatic microclimate (Correia *et al.*, 2019; Da Silva Junior *et al.*, 2021).

The removal of riparian vegetation has serious consequences for lotic ecosystems, such as changing the physicochemical characteristics of water, intensifying erosion processes, and reducing habitat availability, which result in the loss of biological diversity (Siegloch *et al.*, 2016). These effects highlight the urgent need for the continuous and efficient monitoring of the environmental quality of rivers and streams.

In Brazil, studies aimed at monitoring aquatic ecosystems remain insufficient, largely because of the high cost and complexity of conventional methods, which require specialized labor and sophisticated equipment (Minatti-Ferreira & Beaumord, 2006). The lack of public policies and integrated management strategies also contributes to the fragility of conservation actions in these environments.

As viable alternatives, rapid, low-cost, and easily applicable assessment protocols have emerged to support management and conservation strategies (Rigotti *et al.*, 2016). Among these, the Habitat Integrity Index (HII) stands out, having been developed and adapted for Brazilian streams by Nessimian *et al.* (2008). The HII provides qualitative indicators of the degree of environmental conservation based on visual observations of the channel and riparian vegetation attributes, thereby enabling the identification of priority areas for management and restoration (Santos *et al.*, 2020).

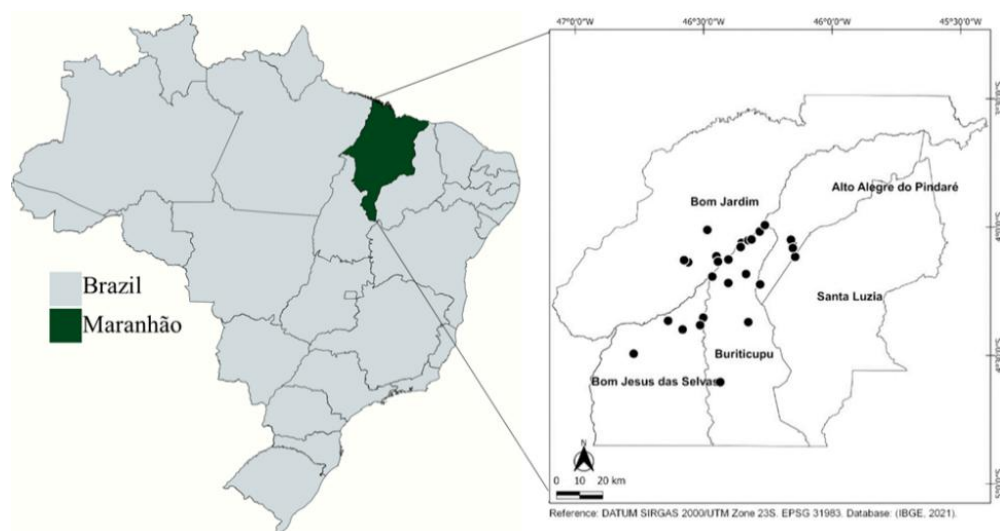
In this context, this study aimed to evaluate the conservation levels of riparian forests and water quality in streams of the Pindaré River Basin, Maranhão, Brazil, by applying the HII and associating it with the measurement of physicochemical variables. This integrated approach contributes to regional-scale environmental monitoring and provides technical support for the conservation and restoration of Amazonian aquatic ecosystems.

MATERIAL AND METHODS

Study Area

In this study, 27 streams were sampled from the Pindaré River Basin and its tributaries located in the municipalities of Alto Alegre do Pindaré, Bom Jesus das Selvas, Buriticupu, Bom Jardim, and Santa Luzia in the state of Maranhão, Brazil (Figure 1). The sampling was conducted between July and November 2022. The annual mean temperature in the study area ranges from 25 to 27 °C. The rainy season is from January to June, and the dry season is from July to December.

Figure 1. Spatial distribution of the 27 streams sampled in the Pindaré River Basin, Maranhão, Brazil



The main economic activity in the basin is livestock farming, which, together with urban area expansion near riverbeds, represents the main anthropogenic factors reducing riparian vegetation and directly contributing to river siltation and erosion (Nicasio *et al.*, 2019).

Measurement of Environmental Variables

For biotic and abiotic characterization, a 100-m stretch was delineated in each of the 27 streams, subdivided into 20 segments of 5 m to measure habitat physical parameters. The HII

was applied, and the riparian forest vegetation, mean canopy cover, channel characteristics, and mean width and depth were measured for each sampled stream.

The HII, developed and adapted by Nessimian *et al.* (2008), analyzes various environmental aspects considering 12 questions that assess: 1 - Land use pattern beyond the riparian zone; 2 - Width of the riparian forest; 3 - Integrity of the riparian forest; 4 - Vegetation of the riparian zone 10 m from the channel; 5 - Retention devices; 6 - Channel sediments; 7 - Bank structure; 8 - Bank erosion; 9 - Watercourse bottom; 10 - Rapids and pools, or meanders; 11 - Aquatic vegetation; 12 - Debris. Each item is scored on a scale of four to six points, representing the lowest to highest level of environmental integrity. The final index is the average value of all habitat characteristics sampled together, ranging from 0 to 1; the closer to 1, the more intact the environment.

This index has been used in several studies, demonstrating its efficiency in assessing environmental changes and their effects on aquatic insect communities (see Brasil *et al.*, 2020). The HII values of degraded environments range between 0.52 and 0.67, altered environments between 0.68 and 0.82, and preserved environments between 0.83 and 1.0 (Souza *et al.*, 2011).

In addition to the HII, the following variables were measured: conductivity, turbidity, dissolved oxygen concentration, pH, temperature, depth, width, and canopy cover. Stream width and depth were measured according to the protocol described by Barbosa-Santos *et al.* (2024). Canopy cover was estimated using photographs captured with the Canopy App (version 1.0.4), which can automatically calculate the canopy percentage from images. The conductivity, turbidity, dissolved oxygen concentration, pH, and temperature were measured using a multiparameter probe (Horiba U-50). All variables were measured at three points (downstream, intermediate, and upstream) along the 100-m stretch, as suggested by Barbosa-Santos *et al.* (2025).

Data Analysis

To compare the differences in the HII values among the different stream categories (preserved, altered, and degraded), the Kruskal–Wallis test was applied, followed by Dunn post hoc tests to check for specific differences. The same analysis was used to assess the differences in physicochemical variables among the categories.

To evaluate the relationship between the HII and environmental variables, a generalized linear model was fitted. The independent variables were electrical conductivity ($\mu\text{S}/\text{cm}$), turbidity (NTU), dissolved oxygen concentration (mg/L), pH, mean depth (cm), mean width (m), water temperature ($^{\circ}\text{C}$), and canopy cover (%). Meanwhile, the HII was considered a continuous response variable.

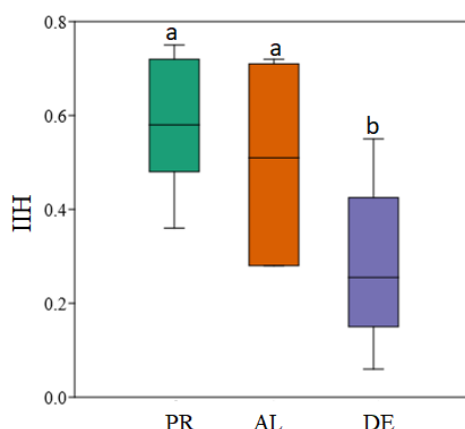
Prior to analysis, the variables were checked for outliers and normality of distribution. Multicollinearity was verified using the variance inflation factor, retaining only variables with values <10. The model was adjusted using the identity link function and the normal distribution of residuals.

Slope coefficients, standard errors, Pearson's correlation coefficients (r), and significance values (p) were calculated for each independent variable.

RESULTS

Of the 27 monitored streams, the HII classified ten as preserved (hereafter, PR), seven as altered (hereafter, AL), and ten as degraded (hereafter, DE). The HII values were higher in PR streams (mean: 0.63 ± 0.11) and lower in DE streams (mean: 0.28 ± 0.16). The Kruskal–Wallis test showed significant differences in the HII values among the streams ($X^2 = 13.25$, $p = 0.001$). Dunn's post hoc test indicated no differences between PR and AL ($p > 0.05$), but significant differences were observed between PR and DE as well as AL and DE ($p < 0.05$) (Figure 2).

Figure 2. Boxplots of the HII values of the target streams: preserved (PR), altered (AL), and degraded (DE). Values followed by the same letter are not significantly different according to Dunn's post hoc test



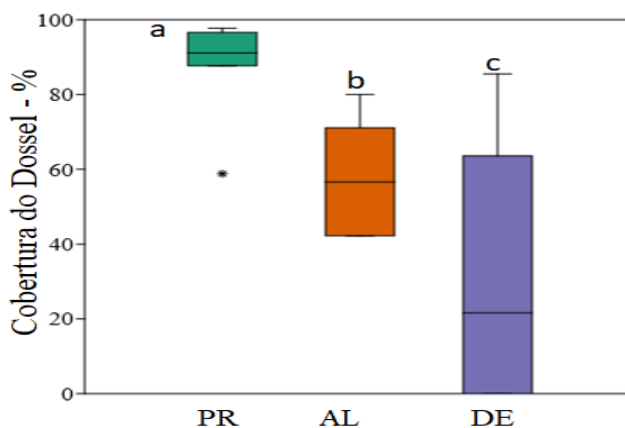
Preserved streams were characterized by high canopy cover, intact riparian vegetation, submerged habitat complexity, and absence of bank erosion (Figure 3A–B). In contrast, fire incidence, riparian vegetation removal, and bank erosion were the main attributes observed in degraded streams (Figure 3C–D).

Figure 3. Environmental characteristics of monitored streams: (A, B) canopy cover and preserved riparian vegetation; (C) presence of siltation; and (D) absence of riparian vegetation



The canopy cover ranged from 0% to 97% in the study area (mean: 58.41 ± 34.78). The highest canopy cover value was recorded in a PR stream (97%) and the lowest in a DE stream (0%), with significant differences among the three categories ($X^2 = 19.29$, $p < 0.0001$) (Figure 4). Dunn's paired analysis indicated significant differences between PR and AL, PR and DE, and AL and DE ($p < 0.05$) (Figure 4).

Figure 4. Boxplots of the canopy cover values of the target streams: preserved (PR), altered (AL), and degraded (DE). Values followed by the same letter are not significantly different according to Dunn's post hoc test

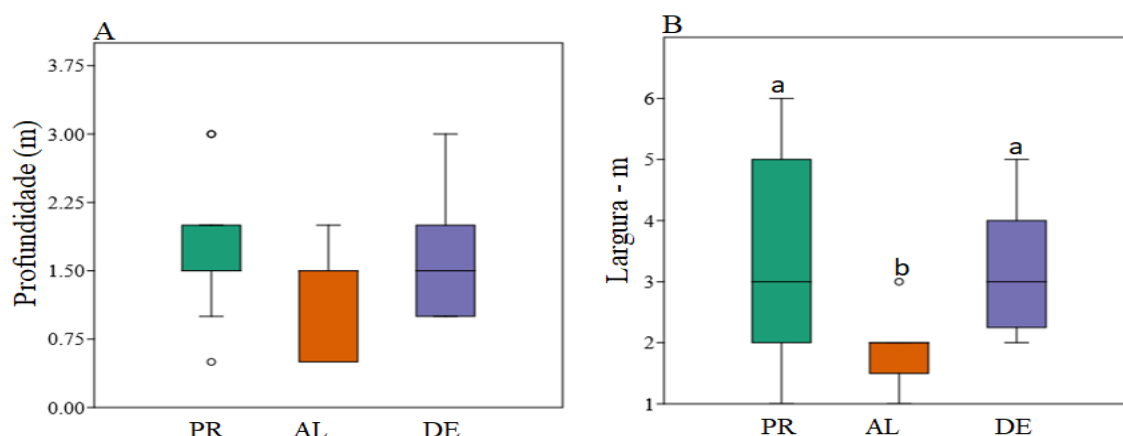


The stream depth varied from 0.50 to 3.00 m (mean: 1.58 ± 0.67), with the greatest depth recorded in a PR stream (3.00 m) and the lowest in an AL stream (0.50 m) (Figure 5A–B). According to the Kruskal–Wallis test, the depth did not significantly differ among the categories ($X^2 = 2.52$, $p = 0.25$) (Figure 6A). In contrast, the stream width differed significantly ($X^2 = 6.08$, $p = 0.037$). Dunn's test showed significant differences between PR and AL and AL and DE ($p < 0.05$), but not between PR and DE ($p > 0.05$) (Figure 6B).

Figure 5. (A) Narrow shallow stream and (B) wide deep stream

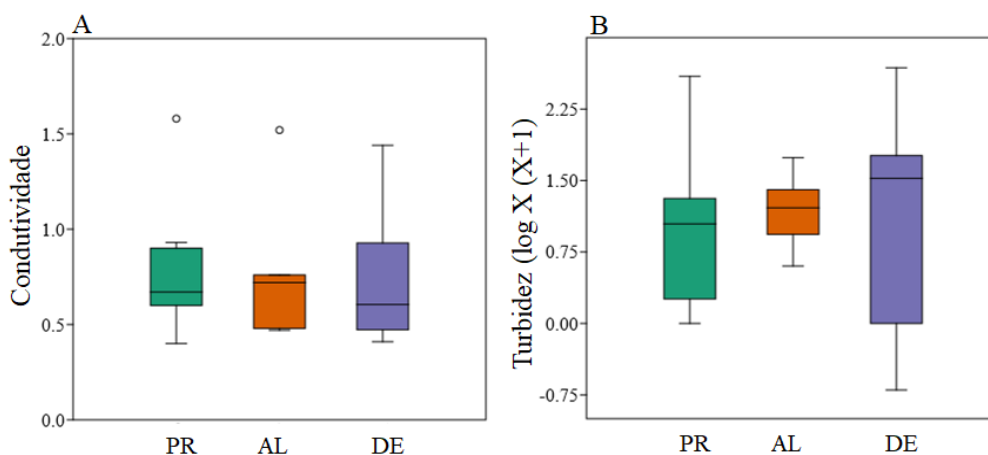


Figure 6. Boxplots of the (A) depth and (B) width of the target streams: preserved (PR), altered (AL), and degraded (DE). Values followed by the same letter are not significantly different according to Dunn's post hoc test



Conductivity ranged from 0.40 to 1.58 (mean: 0.75 ± 0.33). The highest value was observed in a PR stream (1.58) and the lowest in an AL stream (0.43). No significant differences were observed among the categories ($X^2 = 0.45$, $p = 0.79$) (Figure 7A). Turbidity ranged from 0.00 to 482.70 (mean: 51.93 ± 110.23), with the highest values for DE, but without significant differences ($X^2 = 0.31$, $p = 0.85$) (Figure 7B).

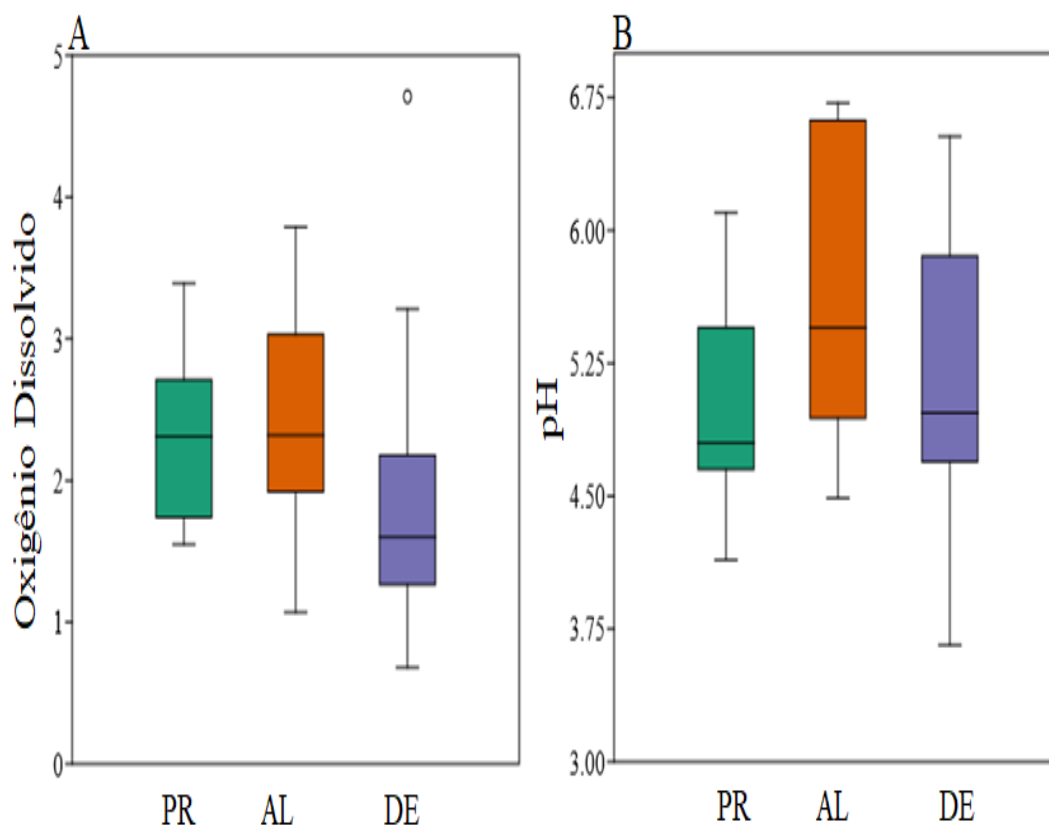
Figure 7. Boxplots of the conductivity (A) and turbidity (B) of the target streams: preserved (PR), altered (AL), and degraded (DE)



The dissolved oxygen concentration ranged from 0.68 to 4.71 (mean: 2.16 ± 0.86). Both the highest and lowest values were observed in DE streams (4.71 and 0.68, respectively). No

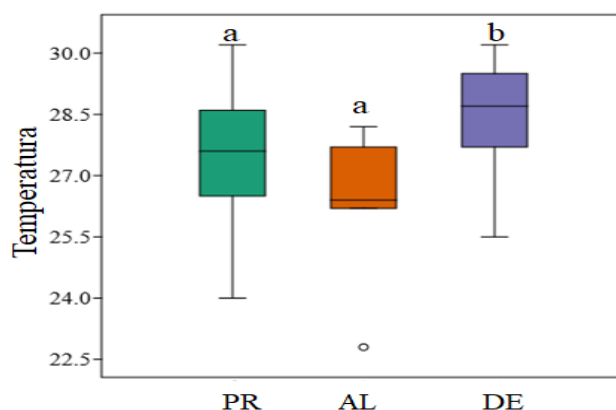
significant differences were found among the categories ($X^2 = 3.8$, $p = 0.14$) (Figure 8A). The pH ranged from 3.66 to 6.72 (mean: 5.2 ± 0.78), with the highest value observed in an AL stream (6.72) and the lowest in a DE stream (3.66), also without significant differences ($X^2 = 2.13$, $p = 0.34$) (Figure 8B).

Figure 8. Boxplots of the (A) dissolved oxygen concentration and (B) pH of the target streams: preserved (PR), altered (AL), and degraded (DE)



The water temperature ranged from 22.80 to 30.20 °C (mean: 27.62 ± 1.74), with the highest temperature recorded in a DE stream (30.2 °C) and the lowest in a PR stream (22.80 °C). Significant differences were detected among the categories ($X^2 = 6.41$, $p = 0.04$). Dunn's test revealed significant differences between PR and DE as well as AL and DE ($p < 0.05$), but not between PR and AL ($p > 0.05$) (Figure 9).

Figure 9. Boxplots of the water temperature of the target streams: preserved (PR), altered (AL), and degraded (DE). Values followed by the same letter are not significantly different according to Dunn's post hoc test



The generalized linear model indicated that only the dissolved oxygen concentration (slope = 1.60; $r = 0.389$; $p = 0.0336$) and canopy cover (slope = 117.38; $r = 0.713$; $p < 0.0001$) had significant positive relationships with the HII, showing that higher values of these variables were associated with better habitat integrity. The other variables analyzed (conductivity, turbidity, pH, depth, width, and temperature) showed no significant correlations ($p > 0.05$), although turbidity and temperature exhibited negative trends against the HII (Table 1).

Table 1. Generalized linear model results between the Habitat Integrity Index (HII) and environmental variables in the target streams. The slope, standard error, intercept, correlation coefficient (r), and significance level (p) are shown

Variable	Slope	Error	Intercept	Error	r	p
Conductivity	0.056	0.297	0.726	0.146	0.03	0.85
Turbidity	-92.683	97.068	93.263	47.755	-0.17	0.34
Dissolved oxygen	1.601	0.716	1.446	0.352	0.38	0.03
pH	-0.230	0.696	5.301	0.342	-0.06	0.74
Depth (cm)	0.046	0.6	1.562	0.295	0.01	0.93
Width (m)	0.078	1.124	2.848	0.553	0.01	0.94
Water temperature	-1.883	1.523	28.459	0.749	-0.22	0.22
Canopy cover (%)	117.38	21.834	6.0544	10.742	0.71	0.0001

DISCUSSION

This study revealed that most streams in the Pindaré River Basin have been altered or degraded, reflecting the intensification of anthropogenic pressure in the region. The low HII values in impacted environments are directly associated with riparian vegetation removal, siltation, and bank erosion, which have also been reported to occur in other Amazon and Cerrado regions (Brasil *et al.*, 2020; Fachinelli & Pereira, 2023; Reis *et al.*, 2017). These results confirm that the HII is sensitive to structural habitat alterations and can serve as an efficient tool for assessing the quality of lotic ecosystems under different disturbance levels.

The loss of riparian vegetation has proven to be decisive in degrading environmental integrity, corroborating studies that point to riparian forests as key elements in the physical and biological stability of watercourses (Barbosa-Santos *et al.*, 2025; Santos *et al.*, 2021). Preserved environments showed high canopy cover, greater structural complexity, and stable margins, whereas degraded streams exhibited excessive channel opening, severe erosion, and higher incidences of fires and pastures. Such structural differences imply changes in local microclimates and habitat availability, which directly affect aquatic and terrestrial biodiversity (Barbosa-Santos *et al.*, 2024; Borges *et al.*, 2021).

The analysis of physicochemical variables complemented the HII interpretation, reinforcing the diagnosis of environmental degradation. Although not all parameters showed significant differences, degraded streams presented higher values of temperature, turbidity, and conductivity, consistent with the increased sediment and nutrient inputs from intensive land use (Brigante *et al.*, 2003; Meyer & Franceschinelli, 2011). In particular, elevated temperatures may be associated with an absence of riparian cover, increased solar radiation, and altered thermal stability in these environments (Bezerra & Andrade, 2022).

Despite the relevance of these findings, some limitations must be highlighted. The absence of biotic data (e.g., macroinvertebrates and fish) restricts the ecological integrity assessment to the physical and chemical attributes. Future studies should thus incorporate biological metrics and multivariate approaches to better understand the relationships between habitat degradation and aquatic biodiversity. Long-term monitoring would also allow for an evaluation of ecosystem resilience in response to anthropogenic pressures and seasonal variations in the Amazon region.

In summary, this study showed that the HII, combined with physicochemical variables, is an effective tool for evaluating the environmental quality of tropical streams. Its use can support public conservation policies and guide sustainable management strategies, thereby contributing



to the protection of water resources in watersheds threatened by intensive land use, such as the Pindaré River Basin.

CONCLUSION

This study found that most streams in the Pindaré River Basin are in altered or degraded states owing to riparian vegetation loss, agricultural expansion, and inadequate land use policies. The application of the HII combined with physicochemical measurements proved effective in elucidating the conservation status of lotic environments, underscoring canopy loss, bank erosion, and turbidity and temperature increases as indicators of degradation.

The results confirmed that riparian vegetation integrity is fundamental for maintaining stream environmental quality, which directly influences water physicochemical conditions and, consequently, aquatic biodiversity. The integration of the HII with limnological parameters reinforces its applicability as a low-cost and easily implementable tool for monitoring tropical aquatic ecosystems.

From a management perspective, the high proportion of impacted streams underscores the urgent need for riparian restoration and anthropogenic impact mitigation for water resources. The use of the HII together with biotic methods, such as macroinvertebrates and fish, as bioindicators could increase the robustness of future assessments.

Thus, the HII represents a practical and efficient strategy to support public conservation policies, identify priority areas for restoration, and promote sustainable water resource use in the Amazonian region of Maranhão.

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